Ankit Pal

Full-Stack Software Engineer · Applied AI Engineer

Mumbai, India · **321ankitpal@gmail.com**
linkedin.com/in/ankit-pal-318913288 · github.com/ankitdecodes
Portfolio: **ankitdecodes.github.io/ankit-protfolio**

SUMMARY

Full-stack software engineer at Finqy (full-time since May 2025, after a five-month internship) building FinTech products end to end: NestJS services, payout and wallet systems, Angular and Next.js frontends and Flutter apps. Backend-focused — transactional outboxes, idempotent ledgers, queue-backed processing and tested business logic — with a growing focus on applied AI that can be measured, not just demoed.

EXPERIENCE

Full Stack Developer · Finqy, Mumbai (Full-time) **May 2025 – Present**

- **Payout pipeline:** built a transactional outbox (LeadApproved / LeadUnapproved events with cycle-aware idempotency keys) drained by a MySQL polling worker — `FOR UPDATE SKIP LOCKED` claiming, capped exponential backoff over 8 attempts, stale-row reclaim and critical alerts.
- **Wallet ledger:** separate balance buckets, pessimistic row locks with double-checked unique idempotency keys, and the agent redeem lifecycle (hold → MIS confirm or release → Finance PAID) with role-based access.
- **Q Finance** (live at qfinance.in): OTP/SMS authentication and profile APIs shared by web and app; credit-bureau report module with PDF generation (Puppeteer, HTML templates); CCAvenue payment integration; unit-tested EMI and top-up calculators.
- **Lead management (LMS):** unified multi-product MIS import with upsert, processed as background jobs by a BullMQ/Redis queue processor; Next.js admin upload workflow; JWT auth with role and permission modules.
- **Frontends & mobile:** set up the Angular SSR website for Q Finance and integrated its APIs; built the FinqyHub and CashKey partner apps in Flutter (Provider, Dio, Firebase push notifications).
- **Quality:** added 40 Jest spec files across two NestJS codebases; shipped through Dockerised dev and UAT environments.

NestJS · TypeScript · MySQL · TypeORM · Redis · BullMQ · Angular · Next.js · React · Flutter · Docker · AWS S3

Full Stack Developer Intern · Finqy, Mumbai **Dec 2024 – Apr 2025**

- Built the Personal Finance Manager module and its NestJS REST APIs for lead creation, assignment and status tracking.
- Implemented recommendation logic from income, expenses and eligibility inputs; designed the MySQL schemas behind it.
- Delivered features on the Angular auto-loan operations dashboard: tranche rules, finance report export and an equipment-loan flow.

PROJECTS

FinPilot AI · Independent · Java, Spring Boot **In progress**

Financial operations platform: transaction analytics, anomaly detection and an investigation assistant whose explanations cite the underlying records. Designed evaluation-first: precision and recall at the deployed threshold will gate every change.

Engineering portfolio · React, TypeScript, Tailwind CSS, Vitest, GitHub Actions **2026**

Interactive simulators of my payout outbox, wallet ledger and redeem lifecycle, plus an applied-AI lab (TF-IDF retrieval, embeddings via PCA, groundedness checks, tool calling) computed in the browser. 70+ automated tests; Lighthouse 100 accessibility; CI/CD to GitHub Pages.

Vocalize — real-time speech translation · Python, TensorFlow, Transformers **2025**

Speech-to-speech pipeline: audio capture → speech recognition (wav2vec / DeepSpeech) → Transformer translation → speech synthesis.

SKILLS

Backend	NestJS, Node.js, TypeScript, REST APIs, JWT / OTP auth, RBAC, TypeORM, BullMQ, transactional outbox, Jest
Frontend	React, Next.js, Angular (SSR), JavaScript, Tailwind CSS · **Mobile:** Flutter, Dart
Data	MySQL, Redis, SQL · also: PostgreSQL, MongoDB
AI / ML	Python, scikit-learn, TensorFlow, Transformers (BERT, GPT), RAG & embeddings, AI evaluation, NumPy, Pandas
Cloud & DevOps	Docker, AWS S3, Git, GitHub Actions, Linux · also: Azure, Kubernetes, Nginx
Java	Core Java, data structures & algorithms, Spring Boot (project-level)

EDUCATION

B.E. Data Science · CGPA 7.4/10 **2021 – 2025**

Lokmanya Tilak College of Engineering, Navi Mumbai

Languages: English (professional), Hindi, Marathi

CERTIFICATIONS

Backend Development — PW Skills (Jul 2023)
MySQL — HackerRank (Jun 2023)
AI / Machine Learning (Data Science) — IBM (Dec 2023)